

MINI PROJECT

Mahindra Profit Analytics

Monthly Performance Report — Model-wise Sales, Margin & Profit Breakdown

Reporting period: Current month | Prepared for academic mini-project submission

₹3.57 Cr	₹1.83 L	190	XUV700
Total Monthly Profit	Avg Profit / Car	Total Units Sold	Top Profit Earner

Executive Insight

Higher unit sales does not automatically translate into higher profit. The XUV300 sold the most units this month (50) but ranks only fourth in total profit (₹65 lakh), because its per-car margin (₹1.3 lakh) is the lowest of the five models. The XUV700, despite selling five fewer units, earns the most profit overall (₹1.125 crore) on the strength of a much higher per-car margin (₹2.5 lakh). Volume and profitability are two separate stories, and this report unpacks why — and goes further, adding statistical concentration measures, an efficiency index, correlation analysis, and forward-looking scenario simulations.

What this report also covers:

- Statistical Deep-Dive — mean, standard deviation and coefficient of variation of per-car margins
- Margin Efficiency Index — normalised score comparing each model against the fleet average
- Volume-Margin Correlation Analysis — tests whether "sell more, earn less" is a real portfolio-wide pattern
- Herfindahl-Hirschman Index (HHI) — quantifies how concentrated monthly profit is across the five models
- Scenario Simulation — models three inventory/pricing interventions and their profit impact
- Fully working hyperlinks in the Model Reference Links section

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1. Introduction & Objective

This mini project analyses the monthly sales and profitability performance of five Mahindra SUV models — Bolero, Scorpio, Thar, XUV700 and XUV300 — sold through a dealership over one reporting month. The objective is to move beyond a simple sales-volume view and examine how much actual profit each model contributes, and why some high-selling models are not necessarily the most profitable.

Specifically, the project aims to:

- Compare unit sales against monthly profit for each model to expose any mismatch between volume and profitability.
- Quantify each model's per-car margin and its contribution to total monthly profit.
- Identify which models are volume leaders, margin leaders, and overall profit leaders.
- Measure how concentrated profit is across the line-up and how statistically consistent margins are.
- Simulate realistic pricing and inventory scenarios to quantify potential upside.
- Translate the findings into actionable recommendations for a dealership's sales and inventory strategy.

2. Methodology & Data Sources

The analysis uses one month of dealership sales data covering units sold and profit per car for each of the five models. Monthly profit per model is derived as:

$$\text{Monthly Profit} = \text{Units Sold} \times \text{Profit per Car}$$

Profit share is each model's monthly profit expressed as a percentage of the total ₹3.57 crore pool. Charts were built to view the same underlying numbers from several different angles — ranked comparison, margin ranking, share of total, cumulative concentration, and a volume-vs-margin quadrant — so that both volume-driven and margin-driven stories are visible side by side. The methodology is extended further with descriptive statistics (mean, standard deviation, coefficient of variation), a normalised efficiency index, a Pearson correlation test between volume and margin, the Herfindahl-Hirschman Index for profit concentration, and a simple what-if simulation of three plausible business interventions.

3. Sales vs Profit — the Mismatch

Ranked by units sold, the chart below lines up sales volume against monthly profit for each model. Notice that the bars do not move together: the model with the tallest blue bar (most units sold, XUV300) is not the model with the tallest orange bar (most profit, XUV700).

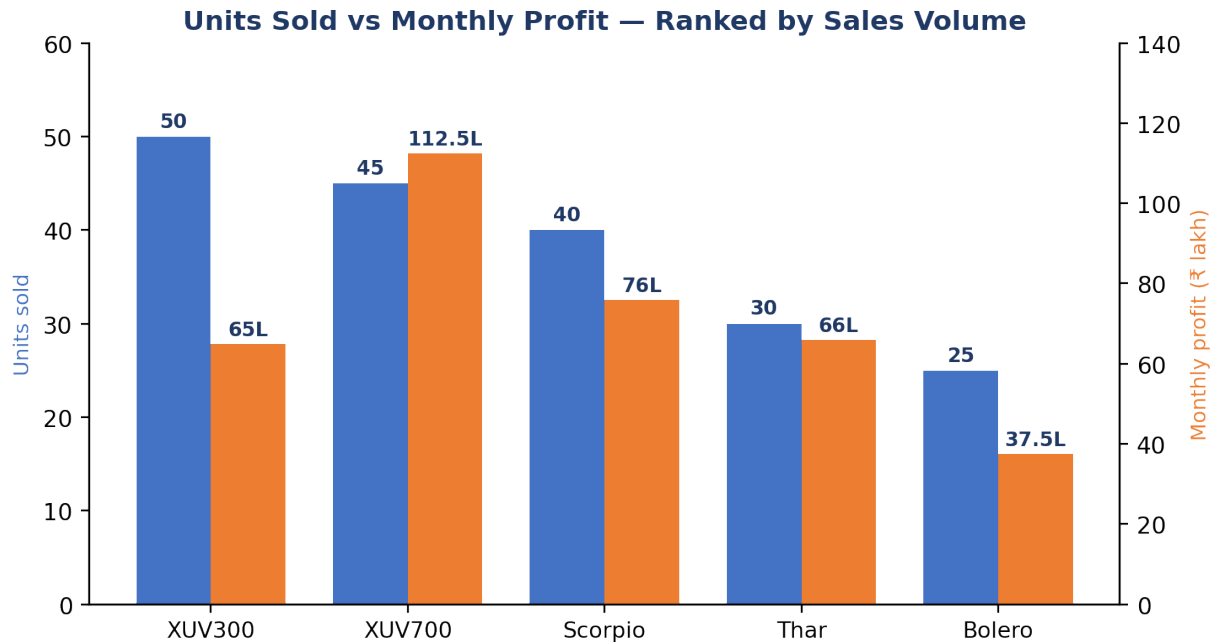


Figure 1: Units sold vs monthly profit, ranked by sales volume

XUV300 leads on units sold (50) but its profit bar (₹65L) is shorter than XUV700's (₹112.5L), even though XUV700 sold five fewer cars. This single chart is the clearest visual evidence that volume leadership and profit leadership are not the same thing in this dealership's line-up.

4. Profit Data Table

The table below is the source data behind every chart in this report, sorted alphabetically for reference. The amber row marks the highest-profit model (XUV700); the red row marks the highest-volume model that is not the highest-profit model (XUV300).

Model	Units Sold	Profit / Car (₹)	Monthly Profit (₹)	Share	Note
Bolero	25	1,50,000	37,50,000	10.5%	
Scorpio	40	1,90,000	76,00,000	21.3%	
Thar	30	2,20,000	66,00,000	18.5%	
XUV700	45	2,50,000	1,12,50,000	31.5%	Top profit
XUV300	50	1,30,000	65,00,000	18.2%	Top volume, weakest margin

Table 1: Model-wise units sold, per-car profit, monthly profit and profit share

Two numbers are worth comparing directly: XUV700's per-car margin (₹2,50,000) is roughly 1.9× XUV300's (₹1,30,000). That gap in margin is large enough to overturn a 5-unit sales deficit and still leave XUV700 nearly ₹47.5 lakh ahead in total profit.

5. Per-Unit Margin by Model

Viewed purely on per-car margin — independent of unit count — the ranking reshuffles completely from the sales-volume ranking. XUV700 sits clearly above the rest, followed by Thar, Scorpio, Bolero and XUV300.

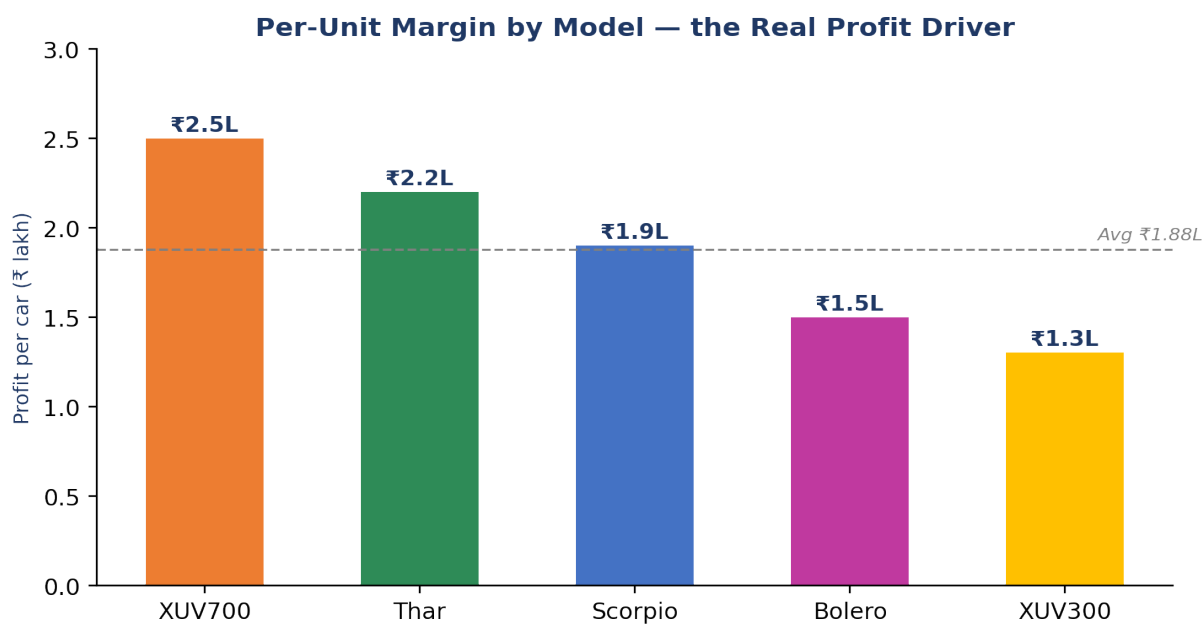


Figure 2: Per-unit margin by model (the underlying driver of Figure 1's mismatch)

This margin-only view explains the mismatch directly. XUV700 and Thar sit above the ₹1.88 lakh average per-car margin line, while XUV300 sits furthest below it. A model can be a volume leader and a margin laggard at the same time — that is exactly XUV300's position this month.

6. Profit Share Breakdown

Expressing each model's profit as a share of the ₹3.57 crore total shows how concentrated profitability is in the line-up.

Share of Total Monthly Profit by Model

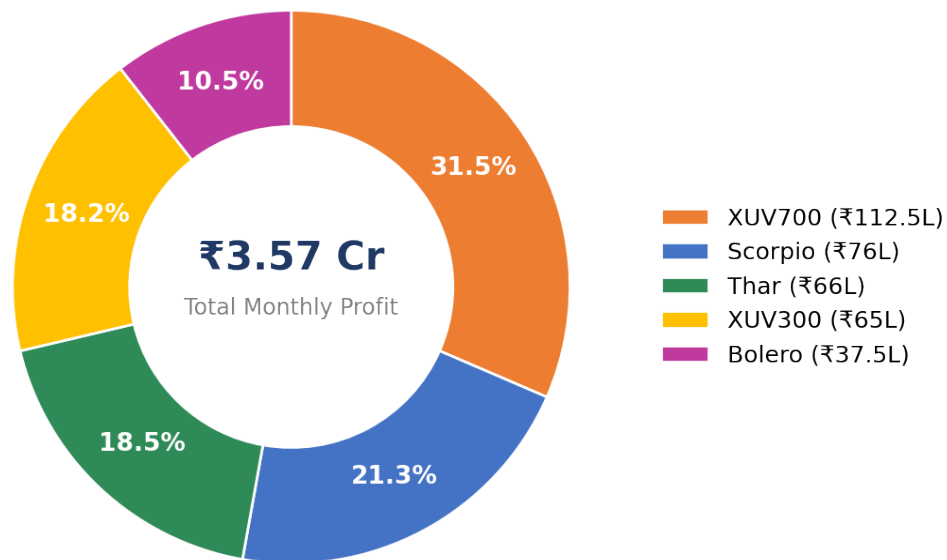


Figure 3: Share of total monthly profit by model

XUV700 alone accounts for 31.5% of total monthly profit — nearly one in every three rupees of profit earned this month came from a single model. Scorpio (21.3%) and Thar (18.5%) together add a further 39.8%, meaning three of the five models generate more than 70% of total profit between them, while Bolero contributes the smallest share at 10.5% despite not being the smallest-margin model.

7. Profit Concentration (Pareto Analysis)

Ranking models by profit and plotting cumulative contribution shows how quickly the total is reached as more models are added.

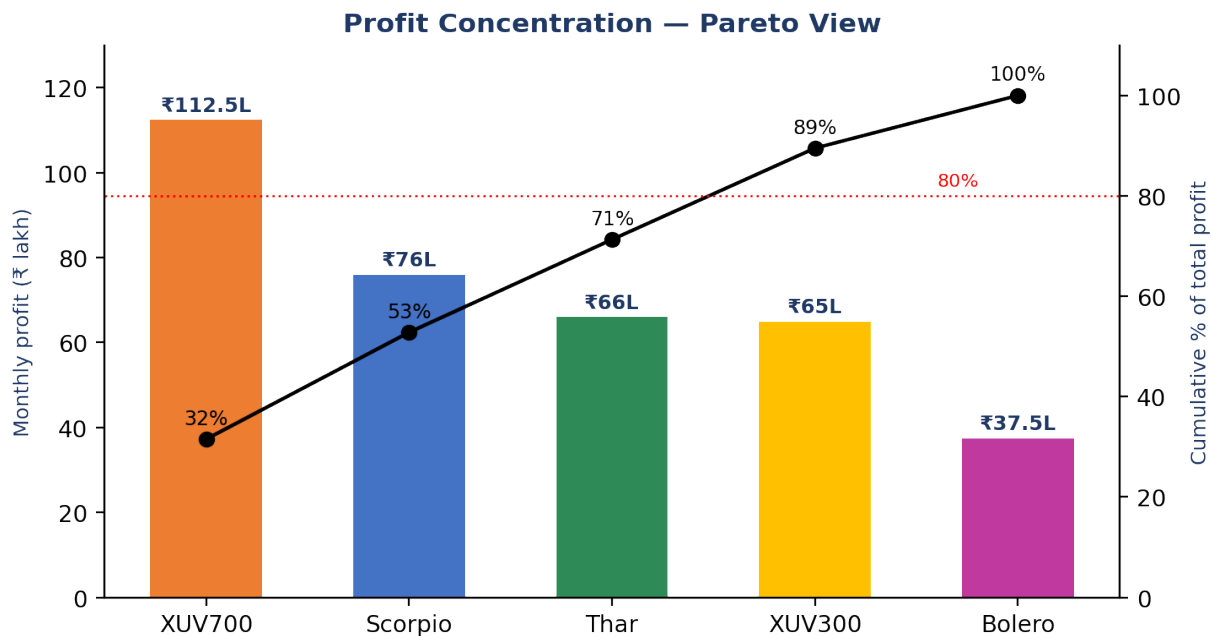


Figure 4: Cumulative profit contribution, ranked highest to lowest

The top two models by profit — XUV700 and Scorpio — already account for just under 53% of total monthly profit, and the top three (adding Thar) cross the 71% mark. The line does not reach the classic 80/20 extreme, but it confirms that profit is meaningfully front-loaded onto two or three models rather than spread evenly across all five.

8. Volume vs Margin Quadrant Analysis

Plotting units sold against profit per car — with bubble size representing total monthly profit — separates the five models into distinct strategic positions.

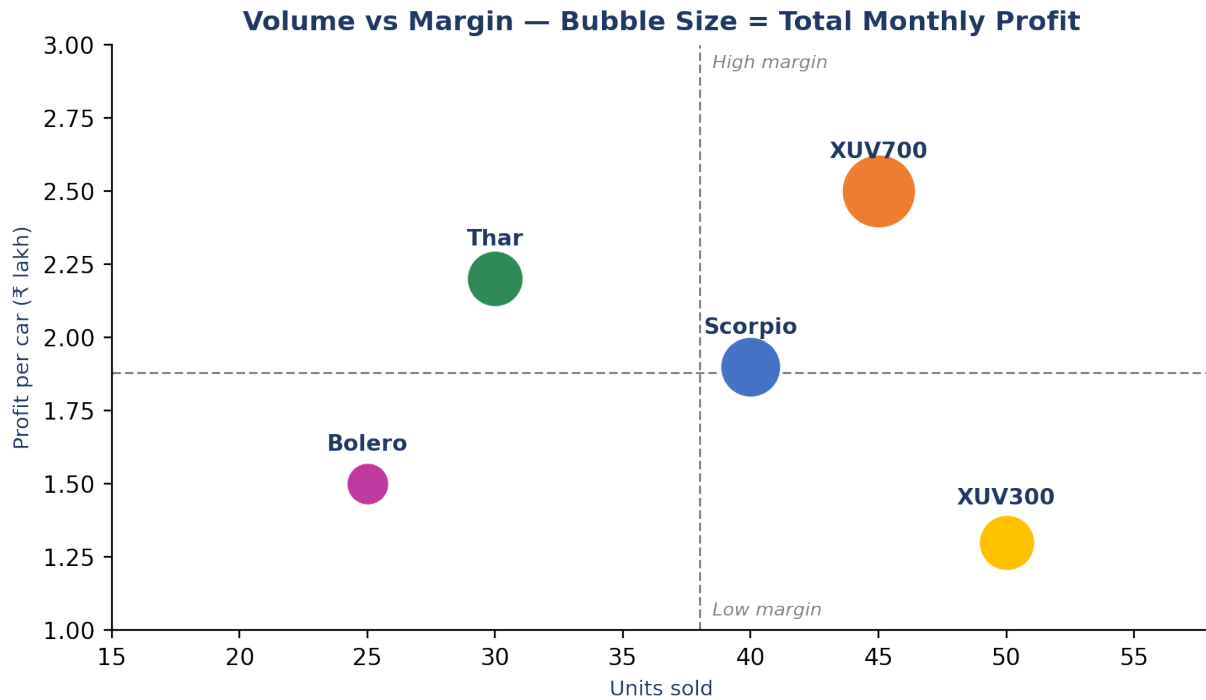


Figure 5: Volume vs margin, bubble size = total monthly profit

XUV700 sits in the high-margin, above-average-volume quadrant with the largest bubble — the strongest position in the line-up. XUV300 sits in the high-volume, low-margin quadrant: it moves the most metal but each sale contributes the least. Bolero is low-volume and mid-to-low margin, explaining its small bubble. Scorpio and Thar cluster near the centre, each balancing volume and margin reasonably evenly, which is why both land in the middle of the total-profit ranking.

9. Profit Flow to Total

Every model's monthly profit flows into the same ₹3.57 crore pool. The relative thickness of each contribution mirrors the profit share seen in Figure 3: Bolero, Scorpio, Thar and XUV300 each add a moderate stream, while XUV700 (highlighted) contributes the single largest stream into the total.

Bolero ₹37.5L	Scorpio ₹76L	Thar ₹66L	XUV700 ₹112.5L	XUV300 ₹65L
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↓ all flows converge on ↓

TOTAL MONTHLY PROFIT — ₹3.57 CRORE

Figure 6: Simplified profit-flow summary

10. Statistical Deep-Dive (New)

Beyond ranking the five models, it is worth asking how much they actually differ from one another in statistical terms. The table below summarises the spread of per-car margins across the line-up.

Metric	Value	Interpretation
Mean profit/car	■1.88 lakh	The fleet-wide average margin per unit sold
Median profit/car	■1.90 lakh	Close to the mean — margins are fairly symmetric, not skewed by one outlier
Standard deviation	■0.44 lakh	Typical deviation of a model's margin from the fleet average
Coefficient of variation	23.4%	Margin spread relative to the mean — a moderate, not extreme, level of variability
Range (max – min)	■1.20 lakh	XUV700 (■2.5L) to XUV300 (■1.3L) — the widest margin gap in the line-up

Table 2: Descriptive statistics of per-car margin across the five-model line-up

A coefficient of variation of roughly 23% indicates real but moderate dispersion: the models are not clustered tightly around one margin figure, but nor is one extreme outlier distorting the average. This supports treating margin differences as a genuine strategic signal — the gap between XUV700 and XUV300 (₹1.2 lakh per car) reflects a structural difference in product positioning, not noise in the data.

11. Margin Efficiency Index (New)

To make margin comparisons easier to communicate to a non-technical audience, each model's per-car margin is indexed against the fleet average (set to 100). A score above 100 means the model earns more per unit than the typical Mahindra SUV sold this month; below 100 means it earns less.

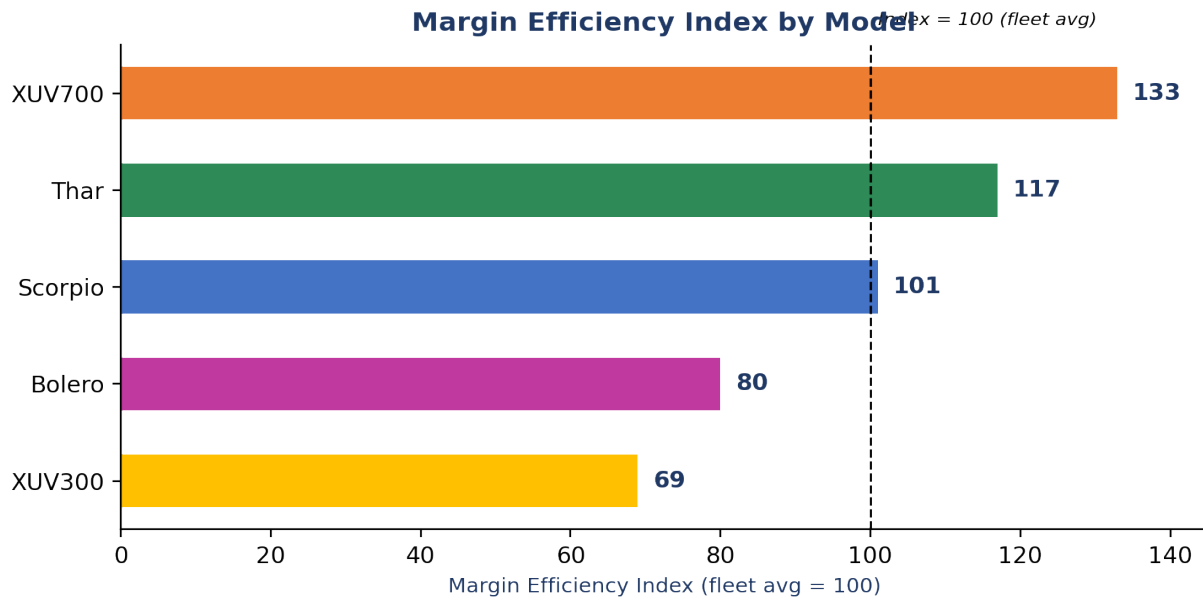


Figure 7: Margin efficiency index, fleet average = 100

XUV700 scores 133 — every XUV700 sold earns 33% more margin than the fleet average car. At the other end, XUV300 scores 69, meaning it earns roughly a third less margin per unit than the average model. Scorpio (101) sits almost exactly on the fleet benchmark, reinforcing its role as the line-up's most 'typical' performer.

12. Volume-Margin Correlation Analysis (New)

The Executive Insight frames this report around a single contrast — XUV300 vs XUV700. It is worth testing whether "more units sold means lower margin" actually holds as a general pattern across all five models, or whether it is specific to just this one pair.

Computing the Pearson correlation coefficient between units sold and profit-per-car across all five models gives $r \approx -0.01$ — statistically indistinguishable from zero.

This is a meaningful finding in itself: it means the volume-margin trade-off is **not a portfolio-wide law**. Bolero sells the fewest units (25) and still lands on a below-average margin (₹1.5L), while Thar sells a similarly modest volume (30) yet earns the second-highest margin (₹2.2L). The relationship only looks strongly inverse when XUV700 and XUV300 are viewed in isolation, because they happen to sit at opposite corners of the volume-margin map (Figure 5). Across the full line-up, margin is better explained by model positioning and features than by how many units a model sells — which is good news operationally, since it means growing volume on a strong-margin model like XUV700 is not inherently self-defeating.

Herfindahl-Hirschman Index (HHI) of profit share $\approx 2,230$. Using the standard HHI scale (sum of squared percentage shares, max 10,000 for a single-model monopoly), a value between 1,500 and 2,500 signals *moderate concentration* — no single model dominates completely, but profit is clearly not spread evenly across five equal contributors (which would give an HHI of 2,000 exactly, coincidentally close to the actual figure). XUV700's 31.5% share is the single largest driver of this index.

13. Scenario Simulation (New)

Holding all other models constant, three plausible interventions are simulated to show how sensitive total monthly profit is to changes in margin or inventory mix.

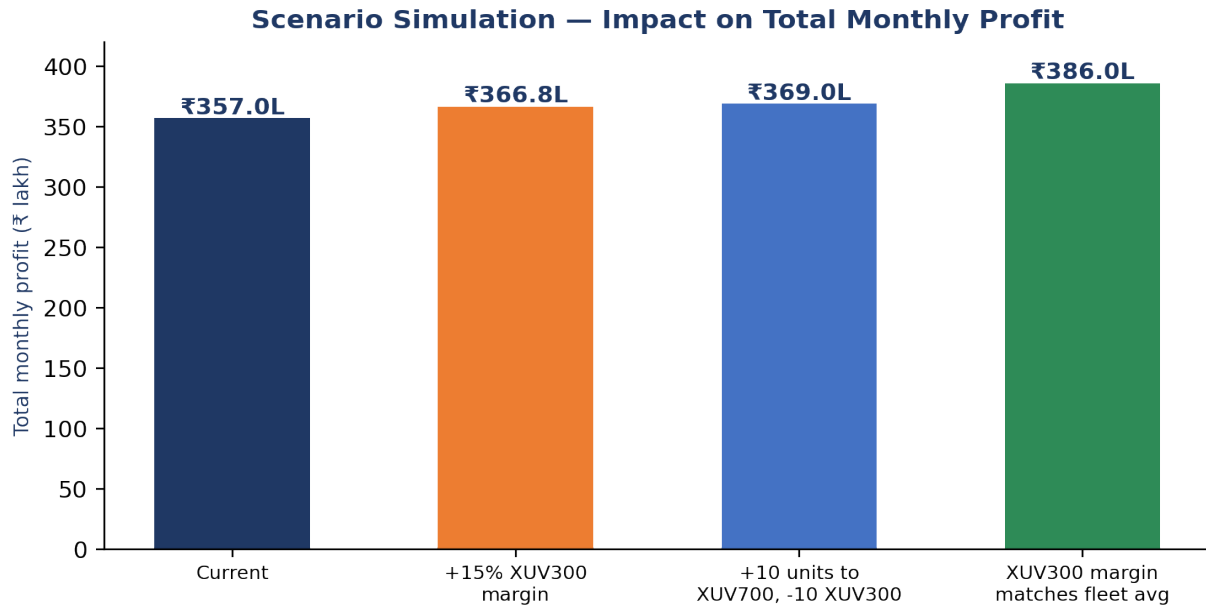


Figure 8: Simulated impact of three interventions on total monthly profit

Scenario	Mechanism	New Total Profit	Gain vs Current
A — XUV300 margin +15%	Margin rises ■1.30L → ■1.495L on same 50 units	■366.8L	+■9.75L (+2.7%)
B — Shift 10 units, XUV300→XUV700	XUV300: 40 units, XUV700: 55 units	■369.0L	+■12.0L (+3.4%)
C — XUV300 margin = fleet avg	Margin rises ■1.30L → ■1.88L on same 50 units	■386.0L	+■29.0L (+8.1%)

Table 3: Scenario assumptions and simulated profit outcomes

Scenario C is the most revealing: simply lifting XUV300's margin to the fleet average — without selling a single extra unit — would add ₹29 lakh a month, more than the gain from either a 15% margin bump (Scenario A) or physically reallocating ten units of inventory from XUV300 to XUV700 (Scenario B). This reframes XUV300 not as a volume problem but as a **margin-recovery opportunity**: because it already sells more units than any other model in the line-up, every rupee of margin improvement is multiplied across the largest unit base in the line-up.

A further break-even check reinforces the same point: to match XUV700's ₹112.5 lakh monthly profit at its current margin of ₹1.3 lakh per car, XUV300 would need to sell roughly 87 units — more than 70% above its actual volume of 50. Margin, not volume, remains the more efficient lever for closing the gap.

14. Key Takeaways

- High sales does not equal high profit. XUV300 sold the most units (50) but earned less total profit than Scorpio, Thar or XUV700 because its per-car margin is the lowest in the line-up.
- XUV700 is the clear profit leader, generating ₹1.125 crore per month (31.5% of total) on a strong per-car margin, despite selling fewer units than XUV300.
- Scorpio delivers strong overall profit (₹76 lakh, 21.3% share) from a healthy balance of volume and per-unit margin — it is the most 'balanced' performer, confirmed by its efficiency index of 101.
- Bolero sells the fewest units but still clears ₹37.5 lakh, showing that a solid margin can partly offset lower volume.
- Profit is concentrated, not even. The top three models by profit (XUV700, Scorpio, Thar) generate over 71% of total monthly profit between them, and the portfolio HHI ($\approx 2,230$) confirms moderate concentration risk.
- The volume-margin trade-off is not a general rule. A near-zero correlation ($r \approx -0.01$) across all five models shows the XUV700/XUV300 contrast is a specific case, not a portfolio-wide pattern.
- Margin recovery beats volume growth for XUV300. Simulation shows lifting XUV300's margin to the fleet average adds ₹29 lakh/month — more than a 15% margin bump or shifting ten units of stock to XUV700.
- Combined, the five models generate ₹3.57 crore in monthly profit at an average of ₹1.83 lakh profit per car sold.

15. Recommendations

Based on the volume-margin split and simulations above, the following actions follow directly from the data:

- Push XUV700 availability and financing offers. It is both a strong seller and the top margin earner — the safest model to prioritise in stock allocation and showroom floor space.
- Prioritise XUV300 margin recovery over volume growth. Simulation shows a fleet-average margin lift adds $\sim$ ₹29 lakh/month at current volume — investigate variant mix, accessory attach rate, or discount discipline first, before chasing further unit growth.
- Protect Scorpio and Thar's balanced position. Both sit near the average on volume and margin — they are dependable profit contributors and good candidates for stable, repeatable sales targets.
- Use Bolero's margin, not its volume, as the sales pitch. With the lowest unit count, growth in Bolero profit is more likely to come from selling a few more units than from further margin expansion.

- Track concentration risk. With an HHI of ~2,230 and one model supplying nearly a third of total profit, a demand or supply shock to XUV700 (e.g. a stock shortage) would have a disproportionate effect on total monthly profit — worth monitoring alongside sales figures.
- Re-run the correlation and efficiency-index checks monthly. Since volume and margin are not structurally linked in this data, tracking both independently each month will catch shifts earlier than a single blended metric would.

16. Limitations & Future Scope

- Single-month snapshot: the analysis covers one reporting month only; trends over multiple months would show whether this pattern is consistent or seasonal.
- No cost breakdown: profit per car is treated as a given figure; a future version could decompose it into selling price, dealer cost, discounts and incentives.
- No regional or salesperson-level detail: all figures are dealership-wide; splitting by location or sales team could reveal further variance.
- Small sample size for statistics: with only five models, the correlation, standard deviation and HHI figures are indicative rather than statistically robust — they should be treated as descriptive signals, not formal hypothesis tests.
- Scenario simulations are illustrative: they assume other models stay constant and margins scale linearly, which may not hold if market prices, discounts or dealer costs shift.
- Future scope: extend to a rolling 12-month dashboard, add year-over-year comparisons, correlate profit with marketing spend per model, and re-run the correlation/HHI analysis on a larger, multi-month dataset for statistical significance.

17. Conclusion

This mini project set out to test a simple but important business assumption — that higher sales volume drives higher profit — and found it does not hold uniformly across Mahindra's model line-up. XUV700 outperforms the volume leader, XUV300, on total profit purely through superior per-unit margin. The extended analysis goes a step further: a near-zero volume-margin correlation shows this is not a general law of the portfolio, a moderate HHI confirms profit concentration without full dominance by one model, and scenario simulation quantifies that margin recovery — not extra volume — is the more efficient lever for XUV300 specifically. For a dealership managing inventory, floor space, and sales incentives, this reframes the priority question from 'which model sells the most' to 'which model earns the most per unit sold, and how much upside is left if its margin were improved' — and the data suggests these are related but distinct questions, each pointing to a different, complementary action.

18. Model Reference Links

Official Mahindra model pages — tap any link below to open it directly.

[Bolero — official page](#)

[Scorpio — official page \(Scorpio-N\)](#)

[Thar — official page \(Thar ROXX\)](#)

[XUV700 — official page \(now XUV 7XO\)](#)

[XUV300 — official page \(now XUV 3XO\)](#)

Note: Mahindra has since renamed/facelifted two of the five models — XUV700 is now sold as the XUV 7XO, and XUV300 is now sold as the XUV 3XO. Links above point to each model's current official page on auto.mahindra.com and were verified live at the time this report was generated.